MQTT Distributed IoT System

Overview

This workshop simulates a distributed IoT architecture using MQTT and Node-RED. The system models multiple zones and mobile agents (trucks) that communicate in real time through a message broker.

The objective is to demonstrate event-driven communication and distributed system design using lightweight IoT protocols.

System Description

The system consists of:

- Multiple zones (e.g., Zone Alpha, Zone Beta)
- Trucks that interact with zones
- An MQTT broker (Mosquitto) enabling communication

Each zone publishes its state to specific MQTT topics, while trucks subscribe to relevant topics to receive updates and trigger actions.

Implementation Details

The architecture is built using Node-RED and MQTT nodes:

- MQTT In nodes subscribe to topics such as: ‘zone/+/data‘
- MQTT Out nodes publish truck status updates such as: ‘truck/1/status’
- Function nodes handle logic such as:
    - Truck arrival events
    - Collection operations
    - State updates
- The dashboard displays real-time trucks activity

The Mosquitto broker runs locally and acts as the central communication hub for all messages, which demonstrates how distributed IoT systems can be designed using MQTT, enabling scalable and event-driven communication between independent components